Application of the three-Coulomb-wave model with variable charge (3CWZ) to the single ionization of CH_4 and NH_3 by electron impact

Final report of the research protocol — DRAFT

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2026

Abstract

This report presents the extension of the three-Coulomb-wave model with a position-dependent effective charge (3CWZ) from atomic to molecular targets, and its application to the single ionization of CH_4 and NH_3 by electron impact.

Model.

- The target is described by the one-centre self-consistent-field (SCF) molecular orbitals of Moccia, expanded on Slater-type functions centred on the heavy atom.
- The random orientation of the molecules is averaged analytically with the orthogonality of the Wigner rotation matrices.
- The effective charges are derived from the monopole part of the Hartree potential of the SCF density. A self-interaction correction distinguishes the ejected electron from the incident one.

Implementation. The model was implemented in a parallel Fortran code and validated against the original CH_4 code and against analytic limits.

Comparisons for NH_3:

- triple differential cross sections (TDCS) measured by El Mir *et al.* at $E_s = 500$ eV, $\theta_s = -6°$, $E_e = 74$ eV;
- electron momentum spectroscopy (EMS) profiles at 1200 eV;
- double differential cross sections (DDCS) at $E_i = 500$ eV.

Results.

- The binary peak is well reproduced for the three valence orbitals $3a_1$, $1e$ and $2a_1$.
- The recoil peak is underestimated by a factor 5–10.
- We show that this deficit comes from the ejected-electron wave. The variable-charge Coulomb wave does not solve the one-electron equation in the screened ionic potential, and it misses the strong back-scattering of the NH_3^+ core: short-range phase shifts of up to 2.6 rad, and elastic back-scattering about 100 times the Coulomb value.
- We delimit the kinematic and numerical domain of validity of the model and propose a distorted-wave treatment of the ejected electron.

To complete: Update the abstract with the CH_4 results, the PCI study and the comparison between variable and constant charge (Sections 6.6–6.8).

Résumé

Ce rapport présente l'extension du modèle à trois ondes coulombiennes avec charge effective variable (3CWZ) aux cibles moléculaires et son application à l'ionisation simple de CH_4 et NH_3 par impact électronique.

Le modèle utilise :

- les orbitales moléculaires SCF monocentriques de Moccia ;
- une moyenne analytique sur les orientations ;
- des charges effectives déduites du potentiel de Hartree, avec une correction d'auto-interaction pour l'électron éjecté.

Pour NH_3, le pic binaire est bien reproduit. Le pic de recul est sous-estimé d'un facteur 5 à 10. Nous montrons que ce défaut provient de l'onde de l'électron éjecté, qui ne contient pas la rétrodiffusion par le cœur ionique. Nous proposons une onde distordue pour cet électron.

To complete: Mettre à jour le résumé après les calculs CH_4, PCI et charge constante.

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1 Introduction

The theoretical study of collisions between electrons and matter is one of the fundamental problems of atomic physics.

Applications. Among the possible collision processes, single ionization by electron impact plays a central role in:

- fundamental collision physics [1];
- plasma physics [2];
- living systems [3];
- atmospheric physics [4];
- radiation–matter interaction [5], where low-energy secondary electrons produced by ionizing radiation cause further ionization, excitation and fragmentation of biomolecules.

The many-body problem. In principle the electron–target system is fully described by the many-body Schrödinger equation. Even for simple atoms this equation has no exact analytical solution, because the electron–electron repulsion couples the electronic motions. Since the work of Hartree and Fock, the exact N-body problem has been replaced by an effective problem in which each electron moves in a mean potential created by the nucleus and the other electrons. This picture describes bound states well. Collisions are harder: an ionizing event has three charged particles in the final state (the scattered electron, the ejected electron and the residual ion), all coupled by long-range Coulomb forces. No exact numerical or analytical solution of this problem exists, so approximate models remain necessary both to interpret experiments and to predict cross sections.

$(e, 2e)$ **experiments.** In kinematically complete experiments, the scattered and ejected electrons are detected in coincidence [6, 7]. They measure the triple differential cross section (TDCS), which is the most detailed probe of the ionization dynamics. The TDCS is sensitive to:

- the electronic structure of the target;
- the post-collision interaction (PCI) between the two outgoing electrons;
- the distortion of the electron waves by the target field.

The 3CWZ model. Theoretical descriptions form a hierarchy of approximations (Section 2.3). Within it, the three-Coulomb-wave (3C or BBK) model [15] treats the three two-body Coulomb interactions of the final state on an equal footing. The 3CWZ model [17, 18, 19] replaces the constant charges of the 3C model by a position-dependent effective charge $Z(r)$. This charge goes from the nuclear charge near the nucleus to the ionic charge far from it, which accounts for electronic screening. The model has improved the description of atomic targets.

Molecular targets. Molecules add four difficulties:

- the potential is not spherical;
- the target has several centres;
- the molecular orientation is random and must be averaged;
- the molecular orbitals are complex.

The choice of CH_4 and NH_3 answers both fundamental and applied interests. Both are ten-electron hydrides. Their C–H and N–H bonds are building blocks of biological, atmospheric and astrophysical molecules [22, 23, 24].

- Methane, with tetrahedral symmetry and a nearly spherical potential, is a favourable case. Several multipolar terms vanish by symmetry in the one-centre representation.
- Ammonia has lower symmetry (C_{3v}) and a permanent dipole moment. It is more anisotropic and a more demanding test of the model.

Objectives. The objectives of the protocol were:

O1. adapt the 3CWZ formalism to molecular targets using the one-centre wave functions of Moccia, including the variable charge $Z(r)$ of each molecular orbital;

O2. compute the TDCS for the single ionization of the orbitals of CH_4 and NH_3 for various kinematics;

O3. study the post-collision interaction by comparing results with and without the electron–electron factor C_{12};

O4. validate the molecular 3CWZ results against the available experimental and theoretical data [37, 38, 39, 40, 41].

The expected outcomes were to show what the variable charge contributes and to delimit the domain of validity of the 3CWZ model for electron–molecule collisions.

Plan of the report.

- Section 2: theoretical background on electron–atom ionization and the 3CWZ final state.
- Section 3: extension to molecules (Moccia orbitals, orientation average, transition amplitude).
- Section 4: the variable effective charge.
- Section 5: numerical implementation, validation and numerical domain of validity.
- Section 6: results for NH_3.
- Section 7: origin of the recoil-peak deficit and proposed improvement.
- Section 8: conclusions and perspectives.

Atomic units ($\hbar = m_e = e = 1$) are used throughout unless stated otherwise.

2 Theoretical background

2.1 Kinematics of the $(e, 2e)$ reaction

An electron of momentum $\mathbf{k}_i$ ionizes a target M in its ground state:

$$e^-(\mathbf{k}_i) + M \longrightarrow M^+ + e^-(\mathbf{k}_s) + e^-(\mathbf{k}_e), \tag{1}$$

where $\mathbf{k}_s$ and $\mathbf{k}_e$ are the momenta of the scattered and ejected electrons. Energy and momentum conservation give

$$E_i = E_s + E_e + I_n, \qquad \mathbf{k}_i = \mathbf{k}_s + \mathbf{k}_e + \mathbf{q}, \tag{2}$$

where:

- $E_\alpha = k_\alpha^2/2$ are the kinetic energies;
- I_n is the binding energy of the ionized orbital;
- $\mathbf{q}$ is the momentum taken by the residual ion.

Two combinations govern the dynamics:

$$\mathbf{K} = \mathbf{k}_i - \mathbf{k}_s \quad \text{(momentum transfer)}, \qquad \mathbf{q} = \mathbf{K} - \mathbf{k}_e \quad \text{(recoil momentum)}. \tag{3}$$

In asymmetric kinematics ($E_s \gg E_e$) the TDCS, as a function of the ejection angle, shows two lobes:

- the *binary peak*, near the direction of $\mathbf{K}$. The target electron is knocked out in a single binary collision with the projectile.
- the *recoil peak*, near $-\mathbf{K}$. The ejected electron is in addition back-scattered by the residual ion, which takes the recoil momentum.

2.2 Triple differential cross section

The TDCS is the probability of detecting the scattered electron in $d\Omega_s$ and the ejected electron in $d\Omega_e$ with energy in dE_e [8]:

$$\sigma^{(3)} = \frac{d^3\sigma}{d\Omega_s\, d\Omega_e\, dE_e} = (2\pi)^4 \frac{k_s k_e}{k_i} |T_{fi}|^2. \tag{4}$$

- The factor $k_s k_e / k_i$ is the phase-space density of the two outgoing electrons.
- All the dynamics is contained in the transition matrix element, written in the first Born approximation as

$$T_{fi} = \langle \Psi_f | V_P | \Psi_i \rangle. \tag{5}$$

Here Ψ_i is the initial state (incident wave times bound orbital), Ψ_f is the final state, and V_P is the projectile–target interaction:

$$V_P = -\frac{Z_T}{r_0} + \sum_{j=1}^{N} \frac{1}{|\mathbf{r}_0 - \mathbf{r}_j|}. \tag{6}$$

In this expression, $\mathbf{r}_0$ is the projectile coordinate, $\mathbf{r}_j$ are the coordinates of the N target electrons, and Z_T is the nuclear charge. In the frozen-core, one-active-electron approximation it reduces to

$$V_P = -\frac{1}{r_0} + \frac{1}{r_{01}}, \qquad r_{01} = |\mathbf{r}_0 - \mathbf{r}_1|, \tag{7}$$

where $\mathbf{r}_1$ is the coordinate of the active electron.

When the two outgoing electrons have comparable energies, they cannot be identified. The TDCS then includes the exchange amplitude [13]:

$$\sigma^{(3)} = (2\pi)^4 \frac{k_s k_e}{k_i} \frac{1}{2} \Big[|T_{dir}|^2 + |T_{exc}|^2 + |T_{dir} - T_{exc}|^2 \Big], \tag{8}$$

In T_{exc}, the roles of $\mathbf{r}_0$ and $\mathbf{r}_1$ in the final state are interchanged. For the asymmetric kinematics studied here ($E_s = 500$ eV $\gg E_e$), exchange is negligible and $T_{fi} = T_{dir}$.

To complete: The 2nd-quarter report writes this formula without the factor $\frac{1}{2}$. Check which convention is used in the atomic 3CWZ papers (the factor does not affect relative TDCS).

2.3 Hierarchy of theoretical models

Table 1 summarizes the main models used to describe ionization by electron impact. They are not opposed to each other. They are successive levels of approximation, each describing the interactions of the process more realistically.

Table 1: Theoretical approaches to electron-impact ionization (adapted from the protocol).

Model	Physical principle	Main limitation
PWBA [9]	Incident and outgoing electrons are plane waves, as if they did not feel the target field	Neglects distortion; valid mainly at high energy
1CW [10, 34]	Plane waves for the projectile; Coulomb wave ($Z = 1$) for the ejected electron	Constant charge; no PCI; no projectile distortion
DWBA, M3DW [12, 13, 14]	Electrons described by distorted waves, numerical solutions in the target potential	PCI treated approximately
3C or BBK [15, 16]	The three final-state particles are Coulomb waves; all three Coulomb interactions included	Constant effective charges; screening not represented
3CWZ [17, 18, 19]	As 3C, with a variable effective charge $Z(r)$ that reproduces electronic screening	Higher numerical cost; molecular extension more delicate

2.4 The 3CWZ final state

The 3CWZ model builds the final state as the product of three Coulomb waves, one for each two-body interaction of the exit channel:

1. scattered electron – residual ion: $\phi_s(Z_s, \mathbf{k}_s, \mathbf{r}_0)$;
2. ejected electron – residual ion: $\phi_e(Z_e, \mathbf{k}_e, \mathbf{r}_1)$;
3. scattered electron – ejected electron (PCI): $C_{se}(\mathbf{k}_{se}, \mathbf{r}_{01})$.

$$\Psi_f = \phi_s^-(\mathbf{k}_s, \mathbf{r}_0)\, \phi_e^-(\mathbf{k}_e, \mathbf{r}_1)\, C_{se}^-(\mathbf{k}_{se}, \mathbf{r}_{01}). \tag{9}$$

Each electron–ion wave is the regular Coulomb wave in parabolic coordinates [20, 21]:

$$\phi_\alpha(Z_\alpha, \mathbf{k}_\alpha, \mathbf{r}) = (2\pi)^{-3/2} e^{-\pi\eta_\alpha/2}\, \Gamma(1+i\eta_\alpha)\, e^{i\mathbf{k}_\alpha\cdot\mathbf{r}}\, {}_1F_1\big(i\eta_\alpha; 1; -i(k_\alpha r + \mathbf{k}_\alpha\cdot\mathbf{r})\big), \qquad \eta_\alpha = -\frac{Z_\alpha}{k_\alpha}. \tag{10}$$

This function is the exact solution of the one-electron Schrödinger equation in a pure Coulomb field of constant charge Z:

$$\Big[-\tfrac{1}{2}\nabla^2 - \frac{Z}{r}\Big]\psi(\mathbf{r}) = E\,\psi(\mathbf{r}). \tag{11}$$

The key step of the 3CWZ model with respect to the 3C model is to replace the constant charges by radially variable charges. The Sommerfeld parameter becomes position-dependent:

$$\eta_\alpha(r) = -\frac{Z_\alpha(r)}{k_\alpha}. \tag{12}$$

- The aim is that the wave feels the bare nuclear charge near the nucleus ($r \to 0$) and the residual ionic charge far from it ($r \to \infty$).
- The incident electron moves in the field of the neutral target. Its wave $\phi_i^+(Z_i, \mathbf{k}_i, \mathbf{r}_0)$ uses the neutral-target charge, which vanishes at large distance.

The electron–electron interaction (PCI) is described by the Coulomb factor [15]

$$C_{se}(\mathbf{r}_{01}) = e^{-\pi\eta_{se}/2}\,\Gamma(1+i\eta_{se})\,{}_1F_1\big(i\eta_{se};1;-i(k_{se}r_{01}+\mathbf{k}_{se}\cdot\mathbf{r}_{01})\big), \qquad \mathbf{k}_{se}=\tfrac{1}{2}(\mathbf{k}_s-\mathbf{k}_e), \quad \eta_{se}=\frac{1}{2k_{se}}. \tag{13}$$

The squared normalization factors of the outgoing waves are the Gamow factors

$$N_\alpha = \left|e^{-\pi\eta_\alpha/2}\Gamma(1+i\eta_\alpha)\right|^2 = \frac{2\pi\eta_\alpha}{e^{2\pi\eta_\alpha}-1}, \tag{14}$$

evaluated with the asymptotic charges. They multiply the TDCS.

Asymptotic conditions. The variable charges reach fixed values at large distance:

$$\lim_{r\to\infty} Z_s(r) = 1, \qquad \lim_{r\to\infty} Z_e(r) = 1, \qquad \lim_{r\to\infty} Z_i(r) = 0, \tag{15}$$

so that the waves satisfy the standard Coulomb boundary conditions.

3 Extension to molecular targets

3.1 Difficulties of the molecular case

Compared with atoms, molecules add four difficulties:

- a non-spherical potential;
- several centres;
- the molecular orientation, which must be taken into account;
- the complex structure of the molecular orbitals.

In particular, a standard multi-centre LCAO description of the target leads to multi-centre integrals that are very hard to evaluate in the collision amplitude. Two strategies overcome these difficulties:

- the one-centre expansion (OCE) of Moccia, which restores a formal similarity with the atomic case;
- the Euler-angle description of the molecular orientation.

3.2 One-centre SCF molecular orbitals of Moccia

Moccia [25, 26] obtained SCF molecular orbitals for XH_n molecules with all basis functions centred on the heavy atom X:

$$\psi_{\mathrm{MO}}(\mathbf{r}) = \sum_j c_j\, f_{n_j l_j m_j}(\zeta_j,\mathbf{r}), \qquad f_{nlm}(\zeta,\mathbf{r}) = \frac{(2\zeta)^{n+1/2}}{\sqrt{(2n)!}}\, r^{n-1}e^{-\zeta r}\, S_{lm}(\theta,\varphi), \tag{16}$$

where ζ_j is the Slater exponent and c_j the SCF coefficient. The real spherical harmonics are [27]

$$S_{lm} = \begin{cases} \dfrac{1}{\sqrt{2}}\left(Y_{l,-|m|} + (-1)^m Y_{l,|m|}\right), & m>0,\\ Y_{l0}, & m=0,\\ \dfrac{i}{\sqrt{2}}\left(Y_{l,-|m|} - (-1)^m Y_{l,|m|}\right), & m<0. \end{cases} \tag{17}$$

To complete: Check this definition against the convention of Ref. [27] used in the code (sign and phase of the $m<0$ functions); the TDCS does not depend on it.

This choice is decisive for the 3CWZ model for two reasons:

- all the functions are centred on one point, so the multi-centre integrals of the amplitude reduce to one-centre integrals;
- the expansion in spherical harmonics is directly compatible with the partial-wave structure of the amplitude.

NH_3. We use Table I of Moccia II [26]: the ground state of pyramidal NH_3 in its calculated equilibrium configuration ($R_{N-H} = 1.9280$ a.u., $\theta_{HNH} = 108.90°$, total energy $E_T = -55.9748$ a.u.). The ten electrons occupy

$$(1a_1)^2(2a_1)^2(1e)^4(3a_1)^2. \tag{18}$$

- $1a_1$ is the N 1s core.
- The valence orbitals are $2a_1$, 1e and $3a_1$. The $3a_1$ orbital is the highest occupied orbital (lone pair).
- The basis has five s functions, three p functions, two d functions and one f function.

Table 2 lists the basis and the coefficients of the valence orbitals. The weights of the angular parts are:

- $3a_1$: 96.4% p, 3.3% s, 0.04% d;
- 1e: 93.6% p, 5.8% d;
- $2a_1$: 95.8% s, 2.8% p, 0.5% d.

The f parts (0.2–0.9%) are kept in the density but not in the ionization amplitude.

Table 2: Slater basis and coefficients of the valence orbitals of NH_3 (Moccia II, Table I). The 1e component shown is $1e_x$; $1e_y$ has the same radial coefficients with the partner harmonics.

n	l	ζ	$2a_1$ ($m=0$)	$3a_1$ ($m=0$)	$1e_x$	
1	0	11.000	0.01157	0.00605		
1	0	6.400	−0.23268	−0.06461		
2	0	1.750	0.75114	0.24313		
2	0	1.280	0.12576	−0.14177		
2	0	2.560	0.14793	0.07510		
3	2	1.600	−0.07830	−0.01440	−0.18794 ($m=1$)	−0.22929 ($m=-2$)
3	2	2.350	0.00659	−0.00699	0.03710 ($m=1$)	0.05282 ($m=-2$)
2	1	1.340	−0.14357	0.95405	1.00304 ($m=1$)	
2	1	1.990	−0.01826	−0.29504	−0.28579 ($m=1$)	
2	1	2.900	−0.00938	0.40188	0.31169 ($m=1$)	
4	3	2.000	0.04992	−0.04098	−0.04008 ($m=1$)	0.06080 ($m=-2$)
4	3 ($m=-3$)	2.000	−0.08013	0.02420		

CH_4. We use the orbitals of Moccia I [25] with $R_{C-H} = 2.08$ a.u. The $1t_2$ orbital (highest occupied orbital) contains p, d and f parts.

3.3 Molecular orientation and Euler angles

In a gas-phase experiment the molecules are randomly oriented. The collision is, however, most conveniently described in the molecular frame, where the potential and the orbitals have a fixed form. The Euler angles (α, β, γ) connect the two frames:

- *Forward transformation:* for a given orientation, a vector of the molecular frame is rotated into the laboratory frame.

- *Inverse transformation:* the measured momenta, known in the laboratory frame, are rotated into the molecular frame to compute the amplitude.

For a linear molecule two angles fix the axis. For non-linear molecules such as CH_4 and NH_3, the third angle γ (rotation about the principal axis) matters too.

For a fixed orientation, the cross section is fourfold differential:

$$\sigma^{(4)} = \frac{d^4\sigma}{d\Omega_{\text{Euler}}\, d\Omega_s\, d\Omega_e\, dE_e}, \qquad d\Omega_{\text{Euler}} = \sin\beta\, d\alpha\, d\beta\, d\gamma, \tag{19}$$

The measured TDCS is its average over all orientations:

$$\sigma^{(3)}_{\text{molecule}} = \frac{1}{8\pi^2}\int \sigma^{(4)}\, d\Omega_{\text{Euler}}. \tag{20}$$

The average is done analytically with the orthogonality of the Wigner matrices [28, 29]:

$$\frac{1}{8\pi^2}\int d\Omega_{\text{Euler}}\, D^{l}_{\mu m}(\alpha,\beta,\gamma)\, D^{l'*}_{\mu' m'}(\alpha,\beta,\gamma) = \frac{\delta_{ll'}\delta_{mm'}\delta_{\mu\mu'}}{2l+1}. \tag{21}$$

Consequence for the TDCS. The amplitude is linear in the orbital. We split the ionized orbital into its angular parts g:

$$\Phi(\mathbf{r}) = \sum_g R_g(r)\, S_{l_g m_g}(\hat{\mathbf{r}}), \qquad R_g(r) = \sum_{j\in g} c_j N_j r^{n_j-1} e^{-\zeta_j r}. \tag{22}$$

A rotation transforms each $S_{l_g m_g}$ into a combination of $Y_{l_g\mu}$ weighted by $D^{l_g}_{\mu m_g}$. Squaring and averaging with Eq. (21) removes all the cross terms between different (l,m) parts and between different μ. The orientation-averaged TDCS is therefore an exact incoherent sum over channels:

$$\langle\sigma^{(3)}\rangle = (2\pi)^4 \frac{k_s k_e}{k_i}\, N_e N_s N_{se}\, n_{\text{el}} \sum_g \frac{1}{2l_g+1} \sum_{\mu=-l_g}^{l_g} \left|T[R_g Y_{l_g\mu}]\right|^2, \tag{23}$$

where:

- $n_{\text{el}} = 2$ is the number of electrons per spatial orbital;
- $T[\chi]$ is the amplitude with the bound orbital replaced by the channel function χ.

For the degenerate 1e level, the TDCS of one component is multiplied by 2.

3.4 Transition amplitude

With the frozen-core interaction (7), the direct amplitude is

$$T_{dir} = \left\langle \phi_s^-\, \phi_e^-\, C_{se}^- \middle| \frac{1}{r_{01}} - \frac{1}{r_0} \middle| \phi_i^+\, \Phi \right\rangle = T^1 - T^0, \tag{24}$$

$$T^1 = \int \phi_s^{-*}(\mathbf{r}_0)\, \phi_e^{-*}(\mathbf{r}_1)\, C_{se}^{-*}(\mathbf{r}_{01})\, \frac{1}{r_{01}}\, \phi_i^+(\mathbf{r}_0)\, \Phi(\mathbf{r}_1)\, d\mathbf{r}_0 d\mathbf{r}_1, \tag{25}$$

$$T^0 = \int \phi_s^{-*}(\mathbf{r}_0)\, \phi_e^{-*}(\mathbf{r}_1)\, C_{se}^{-*}(\mathbf{r}_{01})\, \frac{1}{r_0}\, \phi_i^+(\mathbf{r}_0)\, \Phi(\mathbf{r}_1)\, d\mathbf{r}_0 d\mathbf{r}_1. \tag{26}$$

These six-dimensional integrals are not separable: the factor C_{se} couples the two electrons through $\mathbf{r}_{01} = \mathbf{r}_0 - \mathbf{r}_1$.

Fourier representation of C_{se}. We use

$$C_{se}^{-*}(\mathbf{r}_{01}) = \frac{1}{(2\pi)^3}\int d\mathbf{p}\, e^{i\mathbf{p}\cdot\mathbf{r}_{01}} \int d\mathbf{r}'\, e^{-i\mathbf{p}\cdot\mathbf{r}'}\, C_{se}^{-*}(\mathbf{r}'). \tag{27}$$

Since $e^{i\mathbf{p}\cdot\mathbf{r}_{01}} = e^{i\mathbf{p}\cdot\mathbf{r}_0}e^{-i\mathbf{p}\cdot\mathbf{r}_1}$, the amplitude becomes a three-dimensional integral over $\mathbf{p}$ of products of one-electron integrals:

$$T^1 = \frac{1}{(2\pi)^3}\int d\mathbf{p}\, \underbrace{\int \phi_s^{-*}\phi_i^{+}\, e^{i\mathbf{p}\cdot\mathbf{r}_0} d\mathbf{r}_0}_{\text{projectile}}\, \underbrace{\int \phi_e^{-*}\,\Phi\, e^{-i\mathbf{p}\cdot\mathbf{r}_1} d\mathbf{r}_1}_{\text{ejected electron}}\, \underbrace{\int e^{-i\mathbf{p}\cdot\mathbf{r}'} C_{se}^{-*}(\mathbf{r}')\frac{1}{r'} d\mathbf{r}'}_{\text{PCI}}, \tag{28}$$

$$T^0 = \frac{1}{(2\pi)^3}\int d\mathbf{p} \int \phi_s^{-*}\phi_i^{+}\frac{e^{i\mathbf{p}\cdot\mathbf{r}_0}}{r_0} d\mathbf{r}_0 \int \phi_e^{-*}\,\Phi\, e^{-i\mathbf{p}\cdot\mathbf{r}_1} d\mathbf{r}_1 \int e^{-i\mathbf{p}\cdot\mathbf{r}'} C_{se}^{-*}(\mathbf{r}')\, d\mathbf{r}'. \tag{29}$$

Separation of the short-range part. The variable charges differ from their asymptotic values only in a finite region around the nucleus. Beyond a radius d, the waves reduce to Coulomb waves with constant charge ($Z = 1$ for the outgoing electrons, $Z = 0$, i.e. a plane wave, for the incident one). Each one-electron integral is therefore written as

$$\int_0^\infty \phi(Z(r))\cdots d\mathbf{r} = \int_{r<d} \big[\phi(Z(r)) - \phi(Z_\infty)\big]\cdots d\mathbf{r} + \int_0^\infty \phi(Z_\infty)\cdots d\mathbf{r}. \tag{30}$$

- The second term, with constant charge, is evaluated analytically [10, 11].
- The first term, the short-range correction confined to $r < d$, is evaluated numerically.
- We use $d = 4$ a.u. Section 4 shows that $Z_{ion}(4) = 1.011$, so the variable charge has reached its asymptotic value there to 1%.

4 The variable effective charge

This section summarizes and applies the derivation of the 3rd-quarter report.

4.1 Definition and limits

A charged particle moving through a molecule does not feel a bare point charge. It feels an effective charge equal to the full nuclear charge at the nucleus, which decreases at large distance as the electrons and the light nuclei come into play. We define it as the total electrostatic potential scaled by r:

$$Z_{\text{eff}}(r) \equiv r\, V_{\text{total}}(r), \qquad V_{\text{total}}(r) = V_{\text{nucleus}}(r) - V_{ee}(r) + V_{\text{nuc},H}(r). \tag{31}$$

For a point charge Q, $rV = Q$ exactly, so departures from a constant measure how far the target is from a point charge. The limits are

$$Z_{\text{eff}}(r \to 0) = Z_N, \qquad Z_{\text{eff}}(r \to \infty) = Z_N - N_e + n_H, \tag{32}$$

where N_e is the number of bound electrons and n_H the number of hydrogen nuclei.

4.2 Hartree potential and monopole reduction

The electronic part is the Hartree potential of the SCF density [30, 31]:

$$V_{ee}(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, d^3r', \qquad \rho(\mathbf{r}') = \sum_{\text{MO}} n_{\text{MO}} |\psi_{\text{MO}}(\mathbf{r}')|^2. \tag{33}$$

The Coulomb kernel is expanded in Legendre polynomials [32]:

$$\frac{1}{|\mathbf{r}-\mathbf{r}'|} = \sum_{L=0}^{\infty} \frac{r_<^L}{r_>^{L+1}} P_L(\cos\gamma). \tag{34}$$

For $L \geq 1$, every Y_{LM} integrates to zero over the full solid angle. The spherical average therefore keeps only the monopole $L = 0$:

$$r\,V_{ee}(r) = 4\pi \int_0^r r'^2 \rho_e(r')\,dr' + 4\pi r \int_r^{\infty} r'\rho_e(r')\,dr'. \tag{35}$$

- The first term is the electronic charge enclosed within r: full screening.
- The second term is the partial screening by the electrons beyond r. It vanishes as $r \to \infty$.

This is the shell-theorem result, now identified as the monopole component of the Hartree potential. It coincides with the definition of the protocol and of the 2nd-quarter report: $U(r) = \frac{1}{4\pi}\int V\,d\Omega = -Z(r)/r$.

4.3 Density from Slater-type orbitals and selection rule

With the orbitals of Eq. (16), the density contains cross terms $f_i^* f_j$. Their angular integral is

$$\int S_{l_i m_i} S_{l_j m_j}\,d\Omega = \delta_{l_i l_j}\delta_{m_i m_j}, \tag{36}$$

so only pairs of functions of the same (l, m) contribute to the spherical average. Keeping the spherically averaged density and keeping the monopole Hartree term are the same approximation.

4.4 Closed-form screening integral

For a pair (i, j) of the same (l, m), write $p = n_i + n_j$ and $c = \zeta_i + \zeta_j$. Its contribution is $c_i c_j N_i N_j\, b(r)$ with

$$b(r) = \int_0^r t^p e^{-ct} dt + r\int_r^{\infty} t^{p-1} e^{-ct} dt = \frac{p!}{c^{p+1}} - J_p(c, r) + r\,J_{p-1}(c, r), \tag{37}$$

$$J_n(\alpha, r) = \int_r^{\infty} t^n e^{-\alpha t} dt = \frac{n!}{\alpha^{n+1}} e^{-\alpha r} \sum_{k=0}^{n} \frac{(\alpha r)^k}{k!}. \tag{38}$$

As $r \to 0$, $b \to 0$: there is no screening at the nucleus. As $r \to \infty$, b saturates: the pair screens fully.

4.5 Light nuclei and final expression

The n_H protons are modelled as a thin uniform spherical shell of charge n_H at the X–H distance a_H [32]:

$$Z_H(r) = n_H \min(r/a_H, 1). \tag{39}$$

This keeps only the spherically averaged ($L = 0$) part of their potential, consistently with the monopole treatment of the electrons. Collecting the three contributions:

$$Z_{\text{eff}}(r) = Z_N - \sum_{\text{MO}} n_{\text{MO}} \sum_{\text{same-}(l,m)\text{ pairs}} c_i c_j N_i N_j\, b(r) + n_H \min(r/a_H, 1). \tag{40}$$

4.6 Incident and ejected electrons: self-interaction correction

The incident electron is distinct from the N_e bound electrons and is screened by the full density:

$$Z_{\text{eff}}^{\text{inc}}(r \to \infty) = Z_N - N_e + n_H = 0. \tag{41}$$

The ejected electron comes from one of the occupied orbitals. Keeping the full density would let it be screened by its own charge, which is the self-interaction error of the Hartree approximation. Removing its own contribution,

$$\rho^{\text{ej}}(\mathbf{r}') = \rho(\mathbf{r}') - |\psi_{\text{ionized}}(\mathbf{r}')|^2, \tag{42}$$

reduces the occupation of the ionized orbital by one. Then

$$Z_{\text{eff}}^{\text{ej}}(r \to \infty) = Z_N - (N_e - 1) + n_H = 1, \tag{43}$$

and $Z_{\text{eff}}^{\text{ej}}(r) \geq Z_{\text{eff}}^{\text{inc}}(r)$ at every r. The scattered electron moves in the field of the same ion and uses the same charge. These are the asymptotic conditions of Eq. (15).

4.7 Results for CH_4 and NH_3

Table 3 gives the parameters of the two molecules. Figure 1 and Table 4 show the charges computed by the code from Eq. (40).

Table 3: Parameters of the effective charge.

	CH_4 [25]	NH_3 [26]
Z_N	6 (C)	7 (N)
N_e	10	10
n_H	4	3
a_H (a.u.)	2.08	1.928
Occupied MOs	$1a_1$, $2a_1$, $1t_2$	$1a_1$, $2a_1$, $1e$, $3a_1$
$Z^{\text{inc}}(\infty)$ / $Z^{\text{ej}}(\infty)$	0 / 1	0 / 1

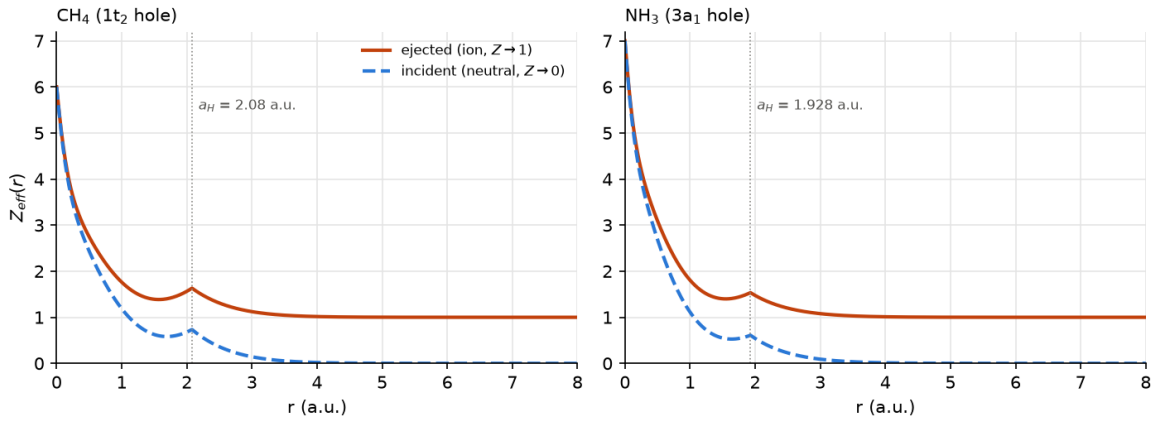


Figure 1: Effective charge of CH_4 ($1t_2$ hole) and NH_3 ($3a_1$ hole): ejected electron (ion, solid) and incident electron (neutral, dashed). The dotted line marks a_H.

Table 4: $Z_{\text{eff}}(r)$ of NH_3 computed by the code.

r (a.u.)	0	0.5	1.0	1.5	2.0	2.5	3.0	4.0
incident (neutral)	7.000	2.690	1.113	0.549	0.541	0.227	0.091	0.013
ejected ($3a_1$ hole)	7.000	3.094	1.806	1.402	1.474	1.198	1.079	1.011

The computed curves show every feature expected from the derivation:

- **Common origin:** $Z(0) = Z_N$ (6 and 7). The fall is steeper for NH_3, whose larger nuclear charge binds the core density more tightly.
- **Fast screening:** by $r = 1$ a.u., Z^{ej} is already 1.8 for NH_3.
- **Local maximum at $r = a_H$:** $Z^{\mathrm{ej}} = 1.53$ and $Z^{\mathrm{inc}} = 0.61$ for NH_3; $Z^{\mathrm{ej}} = 1.63$ and $Z^{\mathrm{inc}} = 0.73$ for CH_4.
  - For $r < a_H$ the rising proton-shell term partly offsets the falling screening term.
  - At $r = a_H$ the proton term stops rising, so Z turns over.
  - In the shell model the maximum is a kink.
- **Asymptotes 0 and 1:** reached by $r \approx 4$ a.u. This justifies the cut-off $d = 4$ a.u. of Eq. (30).
- **Ejected above incident** at every r: this is the signature of the removed self-interaction density.
- The three NH_3 valence orbitals give nearly the same $Z^{\mathrm{ej}}(r)$ (within 0.05).

What the model omits.

- The $L \geq 1$ multipoles of the potential. For NH_3 these include its permanent dipole and the non-spherical field of the protons.
- Exchange and correlation.
- The polarization of the target by the incoming electron.

Z_{eff} is therefore a local, direct, monopole-truncated mean-field potential.

5 Numerical implementation and validation

5.1 Code structure

The code `nh3_opt_claude.for` generalizes the 3CWZ code for the $1t_2$ orbital of CH_4 to any XH_n target described by one-centre Slater orbitals.

- **Target:** one routine contains all the data: nuclear charge, proton shell, all occupied orbitals with their (l, m) parts, occupations of the neutral molecule and of the ion, and the ionized orbital.
- **Effective charges:** computed from these data by Eq. (40) at the radial nodes, for the neutral molecule (incident electron) and for the ion (scattered and ejected electrons).
- **Analytic part:** the $Z = 1$ part of the ejected-electron integral, $\int \phi_e^{-*}(Z = 1)\, R_g Y_{l\mu}\, e^{-i\mathbf{p}\cdot\mathbf{r}} d\mathbf{r}$, is computed in closed form for s, p and d functions of any n and ζ. The part that depends only on $|\mathbf{p}|$ and k_e is computed once per point $\mathbf{p}$.
- **Short-range corrections ($r < d$):** computed by Gauss–Legendre quadrature:
  - projectile: 4 radial intervals × 10 points, and 10×10 points in (θ, φ);
  - ejected electron: 1 radial interval × 10 points.

  A factor $e^{-0.005\, r}$ regularizes the Coulomb integrals.

- **Fourier integral over p:** $60 \times 40 \times 40$ Gauss points, with interval boundaries refined around $|\mathbf{p}| = K$ and around the direction of $-\mathbf{K}$.
- **Mirror symmetry:** in coplanar geometry the wave of the ejected electron is symmetric under $y \to -y$. Combining the $\pm\mu$ channels then halves the angular grid of the ejected-electron correction:

$$A_\mu = \sum_{\text{half}} (c_\mu + (-1)^\mu c_{-\mu}) E^+, \quad B_\mu = \sum_{\text{half}} (c_\mu - (-1)^\mu c_{-\mu}) E^-, \tag{44}$$

$$S_\mu = \tfrac{1}{2}(A_\mu + B_\mu), \quad S_{-\mu} = (-1)^\mu \tfrac{1}{2}(A_\mu - B_\mu). \tag{45}$$

- **Parallelization:** OpenMP over the **p** grid. A coplanar angular distribution of 61 ejection angles takes about 3 min on 4 cores.

5.2 Validation

- **Regression:** with the CH_4 $1t_2$ target, the code reproduces the original CH_4 code to 10 digits.
- **Analytic integrals:** the closed-form $Z = 1$ ejected-electron integrals (s with $n = 1, 2$; p with $n = 2, 3$; d with $n = 3, 4$; all μ) agree with direct numerical integration to 10^{-7}.
- **Symmetry:** the mirror-symmetric path and the full angular grid give the same TDCS to 10 digits (1e, with p and d channels up to $|\mu| = 2$).
- **Target:** all orbitals of Table I are normalized to 1 ± 10^{-5}; $Z(r)$ reaches 0 (neutral) and 1 (ion).

5.3 Additional options

Two options were added to test the physics of the model.

- `isr=1`: the projectile–core interaction $-1/r_0$ of Eq. (7) is replaced by the screened static potential $-Z_{ion}(r_0)/r_0$. This adds its short-range part $-(Z_{ion}(r_0) - 1)/r_0$ for $r_0 < d$, as in the BBKSR model of El Mir *et al.* [33].
- `iorth=1`: the ejected wave is orthogonalized to the ionized orbital (Section 6.3).

5.4 Numerical domain of validity

We repeated calculations with finer grids: 16 points per interval, with 8 radial intervals for the projectile and 4 for the ejected electron.

- **Slow electrons, small** K ($E_e = 74$ eV, $K \approx 0.85$ a.u., the El Mir conditions): the TDCS changes by about 20%. The shapes, and the conclusions drawn from them, are unchanged.
- **Fast ejected electron** (600 eV, EMS kinematics): the TDCS changes by -40% to $+70\%$ along the momentum profile (Table 5). The default single radial interval for the ejected electron cannot resolve a wave with $k_e = 6.6$ a.u. over $r < 4$ a.u.
- **Large momentum transfer** ($E_i = 500$ eV, $E_e = 150$ eV): the default grids overestimate the TDCS by factors 12–30 at $\theta_s = 68°$ ($K \approx 7$ a.u.). At $\theta_s = 120°$ even the finer grid is not stable.

Table 5: Ratio of the TDCS with finer grids to the TDCS with default grids.

EMS, $3a_1$, as a function of the relative azimuth Φ							
Φ	0°	5°	10°	15°	20°	25°	30°
fine/default	0.74	1.30	1.48	1.67	1.53	1.03	0.59
DDCS, 1e, $E_e = 150$ eV, for the five azimuths 0–180°							
$\theta_s = 10°$	1.24, 0.74, 0.89, 1.19, 1.15						
$\theta_s = 68°$	0.08, 0.04, 0.07, 0.03, 0.08						
$\theta_s = 120°$	0.69, 0.10, 0.98, 0.74, 0.88						

The default grids are therefore reliable (about 20%) for slow outgoing electrons and $K \lesssim 3$ a.u. This covers the asymmetric $(e, 2e)$ kinematics of Section 6.1.

To complete: Implement grids adapted to K (radial points scaling with Kd and $k_e d$), or switch to the analytic constant-charge limit beyond $K \approx 3$ a.u., and repeat the convergence study before computing integrated cross sections.

6 Results for NH_3

6.1 TDCS in asymmetric coplanar kinematics

Experimental conditions. El Mir *et al.* [33] measured the TDCS of the three valence orbitals of NH_3 in coplanar asymmetric geometry:

- scattered energy fixed at $E_s = 500$ eV, scattering angle $\theta_s = -6°$;
- ejected energy $E_e = 37$ and 74 eV;
- the incident energy follows from $E_i = E_s + E_e + I_n$.

Table 6 gives the resulting kinematics at 74 eV. The experiment is on a relative scale, so each calculation is normalized to the maximum of the experimental binary peak. The data were digitized from the published figures. There are no experimental points between 160° and 200°.

Table 6: Kinematics at $E_s = 500$ eV, $\theta_s = -6°$, $E_e = 74$ eV.

Orbital	I_n (eV)	E_i (eV)	K (a.u.)	θ_K	$-\mathbf{K}$
$3a_1$	10.85	584.85	0.82	50.2°	230.2°
1e	16.4	590.4	0.84	48.6°	228.6°
$2a_1$	27.0	601.0	0.88	45.7°	225.7°

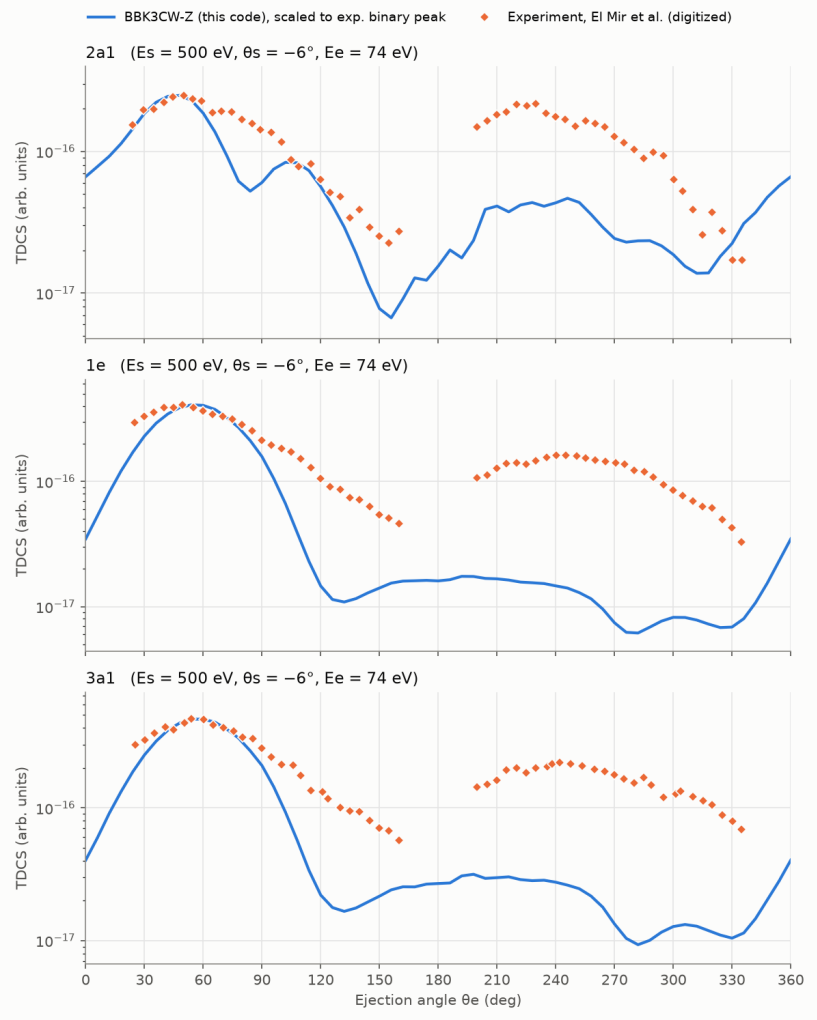


Figure 2: TDCS of the $2a_1$, 1e and $3a_1$ orbitals of NH_3 at $E_s = 500$ eV, $\theta_s = -6°$, $E_e = 74$ eV. Line: 3CWZ, normalized to the experimental binary maximum. Symbols: experiment [33].

Table 7: Peak positions and binary-to-recoil ratio at $E_e = 74$ eV. The recoil maximum is taken in 180–340°.

Orbital		binary peak	recoil peak	binary/recoil
$3a_1$	experiment	54°	242°	2.1
	3CWZ	60°	flat (max. 198°)	14.8
1e	experiment	49°	240°	2.5
	3CWZ	54°	flat (max. 192°)	23.4
$2a_1$	experiment	50°	230°	1.1
	3CWZ	48°	246° (weak)	5.3

Analysis.

- **Binary peak.** Its position is reproduced within a few degrees: 48–60° against 49–54°. The model is slightly too far forward for $3a_1$ and 1e. The front side of the peak (25–60°) follows the data. The Moccia orbitals, the first-order treatment of the fast projectile and the kinematics are therefore adequate.
- **Large-angle side of the binary peak.** The model falls too fast after 90°. At $\theta_e = 120°$, the 1e data are $\approx 1 \times 10^{-16}$ and the model $\approx 1.5 \times 10^{-17}$ (same units after normalization), a factor of 7. The same holds for $3a_1$.

- **Recoil peak.** For $3a_1$ and 1e the model is a flat plateau 5–10 times below the data, with no maximum near $-\mathbf{K}$. For $2a_1$ it shows a weak maximum at 246°, still 4–5 times too low.
- **$2a_1$ structure.** The model has a dip near 80° and a second maximum at 105°, which the experiment does not show. $2a_1$ is mainly N 2s, and its momentum distribution has a node. The model shows this node too sharply.

The first two deficits have one cause: in the model, the ejected electron is not scattered enough by the ion. Every feature away from the binary direction requires the ion to deflect the ejected electron: the slow fall-off, the filling of the valley and the recoil peak.

To complete: Digitize the $E_e = 37$ eV data of [33] and compare with the calculations already done at that energy (files `NH3_*_Es500eV_Ee37eV_*.dat`).

6.2 Short-range projectile–core potential

The frozen-core interaction keeps only the asymptotic charge of the core. With `isr=1` the projectile also feels the screened potential $-Z_{ion}(r_0)/r_0$ inside the molecule. In the amplitude this adds to T^0 a term with the full projectile waves weighted by $(Z_{ion}(r_0) - 1)$, for $r_0 < d$.

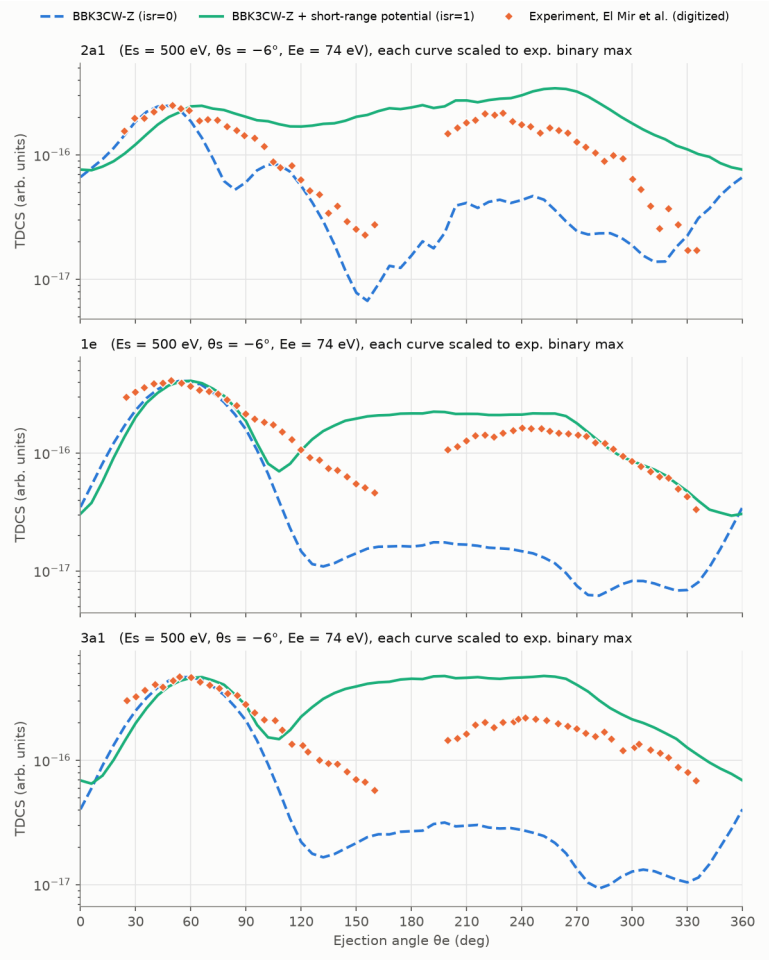


Figure 3: 3CWZ without (dashed) and with (solid) the short-range projectile–core potential. Same kinematics as Fig. 2.

The effect is shown in Fig. 3:

- **1e** gives the best agreement obtained. The binary peak is right and the curve from 270° to 335° falls on the data. However, 130–200° is filled by a plateau that is too high, so the recoil appears as a broad shelf instead of a peak centred at 240°.

- **$3a_1$:** the recoil becomes about 2 times too high and much too broad.
- **$2a_1$:** the binary peak loses its shape and moves to 66°. The curve is almost flat from 60° to 270°.
- **All orbitals:** the binary peak moves forward to 60–66°.

Binary/recoil ratios: 0.98 ($3a_1$), 1.8 (1e), 0.7 ($2a_1$).

The potential produces back-scattering, but nearly isotropic instead of a peak around $-\mathbf{K}$. This behaviour points to the non-orthogonality of the ejected wave to the bound orbital:

- If ϕ_e were an eigenfunction of the same Hamiltonian as Φ, the first-order part of T^0, proportional to $\langle\phi_a|Z/r_0|\phi_i\rangle\langle\phi_e|\Phi\rangle$, would vanish.
- With a Coulomb wave it does not vanish. The short-range term multiplies it by $Z_{ion}(r)-1$, which is up to 6 exactly where the s-like density is concentrated.

6.3 Orthogonalized ejected wave

Method. The standard cure is the Schmidt orthogonalization $\tilde{\phi}_e = \phi_e - \Phi\langle\Phi|\phi_e\rangle/\langle\Phi|\Phi\rangle$. The ejected electron enters the amplitude only through $F(\mathbf{p}) = (2\pi)^{-3/2}\int \phi_e^{-*}\, e^{-i\mathbf{p}\cdot\mathbf{r}}\chi\, d\mathbf{r}$, so the orthogonalized amplitude uses

$$\tilde{F}(\mathbf{p}) = F(\mathbf{p}) - \frac{S\,G(\mathbf{p})}{n}, \qquad S = (2\pi)^{3/2}F(\mathbf{0}), \qquad G(\mathbf{p}) = (2\pi)^{-3/2}\int |\chi|^2 e^{-i\mathbf{p}\cdot\mathbf{r}} d\mathbf{r}, \tag{46}$$

where $n = \langle\chi|\chi\rangle$. Orthogonalizing to the full orbital at each orientation would make the amplitude non-linear in Φ and destroy the analytic average of Eq. (23). We therefore orthogonalize each channel $\chi_{g\mu} = R_g Y_{l_g\mu}$ to itself. This is the atomic prescription applied to every partial wave of the one-centre expansion. With the plane-wave expansion, the density form factor is

$$\frac{G_{g\mu}(\mathbf{p})}{n_g} = \sum_{L=0,2,4} \frac{4\pi(-i)^L}{(2\pi)^{3/2} n_g} A_{l\mu L}\sqrt{\tfrac{2L+1}{4\pi}} \int_0^\infty R_g^2(r) j_L(pr) r^2 dr\; P_L(\cos\theta_p), \tag{47}$$

$$A_{l\mu L} = \int |Y_{l\mu}|^2 Y_{L0}\, d\Omega. \tag{48}$$

A brute-force three-dimensional integration agrees with this formula to 6×10^{-8}.

Overlaps. Table 8 gives $|\langle\phi_e|\chi_{g\mu}\rangle|$ at $\theta_e = 0$. The largest overlaps are in the channels where the short-range term overshot.

Table 8: Overlaps $|\langle\phi_e|\chi_{g\mu}\rangle|$ at $E_e = 74$ eV, $\theta_e = 0$.

Orbital	Overlaps
$2a_1$	s: 0.37; p, d: ≤ 0.02
$3a_1$	s: 0.10; p: 0.18–0.20; d: ≤ 0.01
1e	p: 0.028; d: 0.003–0.03

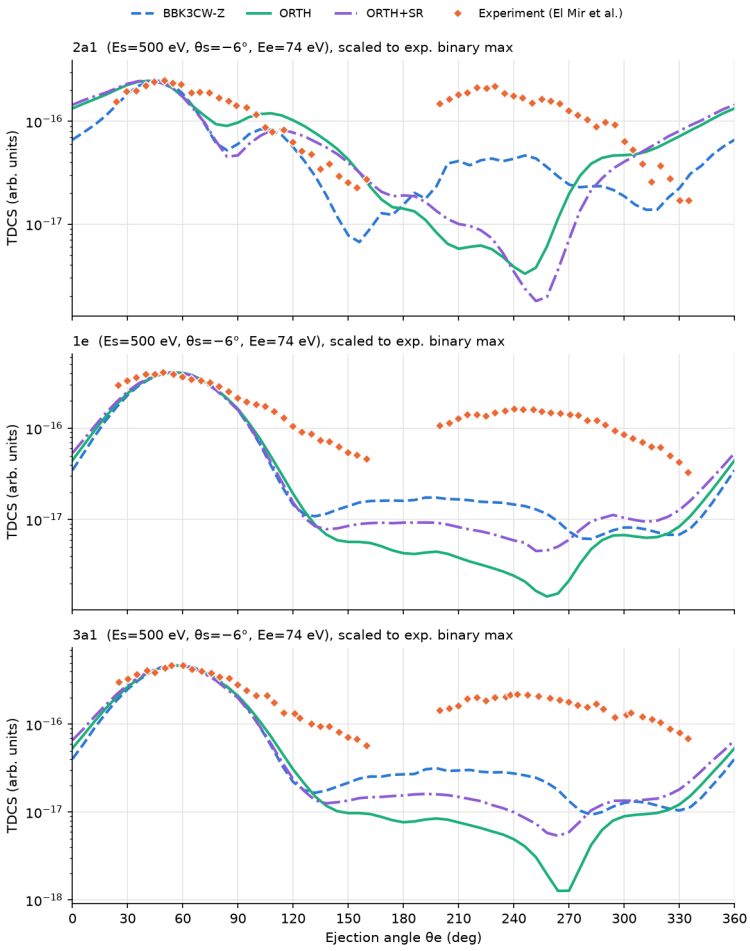


Figure 4: 3CWZ (dashed), orthogonalized ejected wave (solid), and orthogonalized wave with the short-range potential (dash-dot), compared with experiment [33].

Results (Fig. 4, Table 9).

- The binary peak is unchanged.
- Without the short-range term, orthogonalization lowers the recoil region further, by a factor 2–4.
- With it, the isotropic shelf disappears and $2a_1$ recovers its binary shape, but the recoil falls back to about 10 times below the data.

Most of the back-scattering produced by the short-range term was therefore the overlap artefact. The physical back-scattering contained in the 3CWZ ejected wave is too weak.

Table 9: Binary maximum divided by the mean TDCS in 200–300° (lower means a stronger recoil).

Orbital	experiment	3CWZ	+ short-range	orthogonalized	orth. + short-range
$3a_1$	2.6	22	1	91	44
1e	3.0	34	2	117	55
$2a_1$	1.6	7	1	15	20

6.4 Electron momentum spectroscopy

Conditions. Rezkallah *et al.* [34] tested the Moccia orbitals against the EMS measurements of Bawagan *et al.* [35]:

- incident energy 1200 eV+I_n;
- symmetric non-coplanar geometry: both outgoing electrons at $\theta = 45°$ with 600 eV each;
- the relative azimuth Φ is varied, which varies the recoil momentum

$$q = \left[(2k_e \cos\theta - k_i)^2 + 4k_e^2 \sin^2\theta \sin^2(\Phi/2)\right]^{1/2}. \tag{49}$$

In this regime, the TDCS is proportional to the spherically averaged momentum density of the orbital. We ran the 3CWZ code in the same conditions, with $\Phi = 0$–$30°$ in $1°$ steps. The experimental points and the PWBA curves were read from the figures of Ref. [34].

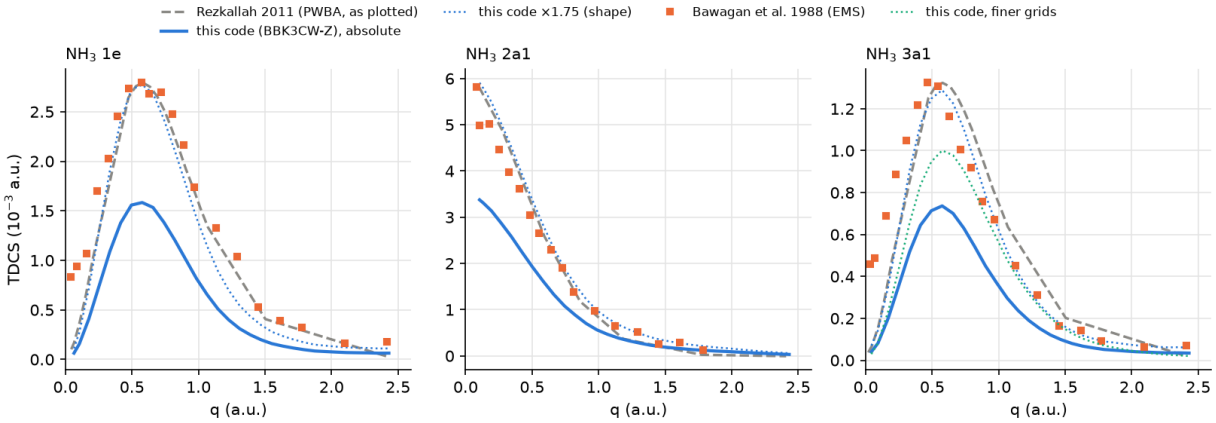


Figure 5: NH_3 momentum profiles at 1200 eV. Lines: 3CWZ, absolute and multiplied by 1.75; finer grids for $3a_1$; PWBA of Ref. [34]. Symbols: experiment [35].

Results (Fig. 5).

- **Peak positions** are reproduced: $q \approx 0.58$ (1e), 0.1 ($2a_1$) and 0.57 ($3a_1$) a.u.
- **Shapes.** After normalization (factor 1.72–1.80), $2a_1$ and $3a_1$ follow the data up to $q \approx 2.4$ a.u.: the ratio experiment/model is 0.9–1.1 for $3a_1$ at $q > 1.3$. For 1e it is 1.3–1.8 at $q > 1.3$ a.u. The PWBA of Ref. [34] shows the same deficit, so the high-momentum content of the 1e orbital is slightly too small.
- **Low q.** For $q < 0.2$ a.u., the data of the p-like orbitals 1e and $3a_1$ are 2–7 times above all calculations. This is the known turn-up of EMS profiles, due to distortion and finite angular resolution.
- **Absolute values.** With the default grids the 3CWZ peaks are 0.56–0.58 of the PWBA for all three orbitals: 1.58, 3.38 and 0.74 against 2.79, 5.82 and 1.32 ($\times 10^{-3}$ a.u.). With finer grids the $3a_1$ peak rises from 0.74 to 1.00. Most of the difference is therefore numerical (Section 5.4).

6.5 Double differential cross section

Method. The DDCS is obtained by integrating the TDCS over the scattered electron:

$$\frac{d^2\sigma}{d\Omega_e\, dE_e} = \int \sigma^{(3)} d\Omega_s = \frac{1}{k_i k_s} \int_{K_{\min}}^{K_{\max}} K\, dK \int_0^{2\pi} \sigma^{(3)}\, d\varphi, \tag{50}$$

- 17 points equally spaced in $\ln K$ between $k_i - k_s$ and $k_i + k_s$;
- azimuths 0–180° in 45° steps, by symmetry;

- Simpson's rule in both variables;
- the three valence orbitals summed, with 1e counted twice.

The result is compared with the data of Opal *et al.* [36] at $E_i = 500$ eV and $\theta_e = 120°$, read from Ref. [34].

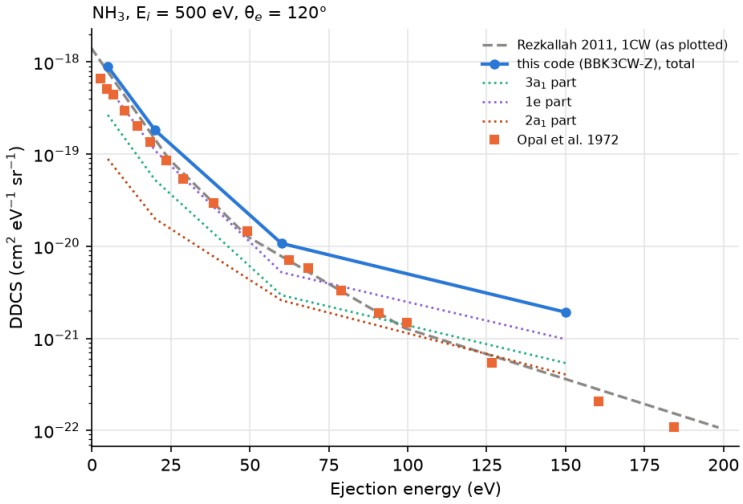


Figure 6: DDCS of NH_3 at $E_i = 500$ eV, $\theta_e = 120°$: 3CWZ (total and per orbital), 1CW of Ref. [34], and experiment [36].

Table 10: DDCS at $E_i = 500$ eV, $\theta_e = 120°$ ($\mathrm{cm^2\,eV^{-1}\,sr^{-1}}$).

E_e (eV)	3CWZ	Opal *et al.*	ratio	share from small θ_s	
5	8.94×10^{-19}	5.00×10^{-19}	1.79	0.82–0.90	($\theta_s < 10$–$15°$)
20	1.82×10^{-19}	1.15×10^{-19}	1.58	0.60–0.89	(< 14–$18°$)
60	1.08×10^{-20}	7.97×10^{-21}	1.35	0.34–0.64	(< 22–$25°$)
150	1.92×10^{-21}	2.71×10^{-22}	7.1	0.26–0.31	(< 35–$37°$)

Results (Fig. 6, Table 10).

- From 5 to 60 eV the model exceeds the data by factors 1.8–1.35. The simpler 1CW model of Ref. [34] is closer to the data.
- At 150 eV the excess (factor 7) comes for 70% from large scattering angles ($\theta_s > 35°$), where the TDCS with the default grids levels off instead of decreasing. The finer-grid tests (Table 5) show that this plateau is numerical.
- Orthogonalizing the ejected wave does not remove it: the ratio goes from 7.1 to 6.4.

To complete: Repeat the DDCS with K-adapted grids, then integrate over Ω_e and E_e (SDCS and total cross section) and compare with Refs. [36, 40].

6.6 CH_4: TDCS

To complete: TDCS of the CH_4 $1t_2$ and $2a_1$ orbitals with the same code, compared with Lahmam-Bennani *et al.* [37] ($E_s = 500$ eV, $E_e = 37$ and 74 eV), and if possible with the low-energy data and M3DW calculations of Refs. [38, 39] (objectives O2 and O4). The CH_4 target is already in the code (switch `iorb=0` for $1t_2$). The $2a_1$ data must be added.

6.7 Post-collision interaction

To complete: Objective O3: rerun the NH_3 orbitals at $E_e = 74$ eV and CH_4 $1t_2$ with $C_{se} = 1$ and $N_{se} = 1$. Report the shift of the binary peak, the change of the binary/recoil ratio, and the effect at $E_e = 37$ eV, where PCI should be stronger.

6.8 Variable versus constant charge

To complete: Expected result 1: rerun with $Z \equiv 1$ for the outgoing electrons and $Z \equiv 0$ for the incident electron (3C/BBK limit) to isolate the effect of the variable charge on the binary and recoil peaks.

7 Discussion: origin of the recoil deficit

7.1 Momenta probed by the experiment

In a plane-wave impulse picture, the TDCS is proportional to the momentum density $\rho(q)$ of the orbital at $q = |\mathbf{K} - \mathbf{k}_e|$. Table 11 gives the values of q at the binary peak, at 120° and at the recoil peak for the El Mir kinematics, and the ratios $\rho(q_{\mathrm{binary}})/\rho(q)$ computed from the Moccia orbitals.

Table 11: Recoil momenta and plane-wave estimates at $E_s = 500$ eV, $\theta_s = -6°$, $E_e = 74$ eV.

Orbital	q_{binary}	$q(120°)$	q_{recoil}	$\rho(q_b)/\rho(q_{120})$	$\rho(q_b)/\rho(q_r)$
$3a_1$	1.51	2.19	3.16	5.6	38
1e	1.49	2.21	3.18	7.3	59
$2a_1$	1.45	2.26	3.22	49	155
Experiment					1.1–2.5

Two conclusions follow.

- The whole measured TDCS lies in the high-momentum tail of the orbitals. The EMS test showed that this tail is somewhat too small for 1e, which may contribute to the narrow binary peak.
- No realistic orbital has as much density at $q \approx 3.2$ as at $q \approx 1.5$ a.u. The measured recoil peak cannot be a structure effect. It must come from the scattering of the ejected electron by the residual ion.

7.2 Exact solution of the one-electron equation

The 3CWZ ejected wave aims to describe the motion in the screened ionic field, Eq. (11) with $Z \to Z_{ion}(r)$. We solved this equation exactly by partial waves:

$$u_l''(r) + \left[k^2 - \frac{l(l+1)}{r^2} + \frac{2Z_{ion}(r)}{r}\right] u_l(r) = 0, \tag{51}$$

with the $Z_{ion}(r)$ of the code ($3a_1$ hole) and $k = 2.33$ a.u. ($E_e = 74$ eV). The solution was integrated to $r = 40$ a.u. and matched to the regular and irregular Coulomb functions F_l, G_l for $Z = 1$:

$$u_l \propto \cos\delta_l \, F_l(\eta, kr) + \sin\delta_l \, G_l(\eta, kr), \tag{52}$$

which defines the short-range phase shifts δ_l beyond the $Z = 1$ Coulomb wave.

Table 12: Short-range phase shifts of $e^- + NH_3^+$ at 74 eV.

l	0	1	2	3	4	5	6	7	8
δ_l (rad)	2.60	1.34	0.48	0.26	0.17	0.09	0.04	0.017	0.007

- $\delta_0 = 2.60$ is equivalent to -0.54 modulo π. The value above $\pi/2$ reflects the bound s states of the potential (Levinson's theorem).
- $\delta_1 = 1.34$ is close to $\pi/2$, the maximum effect: the p wave is the dominant missing channel.
- The shifts fall below 0.2 rad for $l \geq 4$.

Comparison of the partial waves. We projected the 3CWZ wave of Eqs. (10)–(12) on Legendre polynomials at fixed r. A true continuum solution has a constant phase $\sigma_l + \delta_l$ (mod π) in each partial wave. The 3CWZ wave does not. Its phase drifts with r towards the $Z = 1$ value, not towards the exact one (Table 13).

- Its radial nodes are displaced. The exact s wave has a node near $r = 0.3$ a.u., where it must be orthogonal to the 1s-like core: amplitude 0.08 (exact) against 0.56 (3CWZ).
- At $r = 0.6$ a.u. the situation is reversed: 0.53 against 0.09.

Table 13: Phase (mod π, rad) of the partial waves at $E_e = 74$ eV.

l	exact (all r)	3CWZ ($r = 0.3 \to 2.5$ a.u.)	Coulomb $Z = 1$
0	-0.33	$0.07 \to 0.25$	0.22
1	1.15	$-0.96 \to -0.22$	-0.19
2	0.08	$-1.66 \to -0.48$	-0.40
3	-0.28	$-2.17 \to -0.65$	-0.54

Back-scattering. The elastic scattering amplitude of the electron on NH_3^+ follows from the phase shifts:

$$f(\theta) = f_C(\theta) + \frac{1}{2ik}\sum_l (2l+1)e^{2i\sigma_l}\left(e^{2i\delta_l} - 1\right)P_l(\cos\theta), \tag{53}$$

where f_C is the Rutherford amplitude for $Z = 1$. Table 14 compares $|f|^2$ with $|f_C|^2$.

Table 14: Elastic $e^- + NH_3^+$ cross section divided by the Rutherford value for $Z = 1$, at 74 eV.

angle	30°	60°	90°	120°	150°	180°
$\lvert f\rvert^2/\lvert f_C\rvert^2$	2.5	4.8	4.9	29	91	137

- At forward angles the ion potential is only moderately stronger than $Z = 1$.
- At backward angles, the strong field near the nitrogen (Z up to 7) reflects the electron about 100 times more efficiently.

This is exactly the process the recoil peak requires. The 3CWZ ejected wave keeps the $Z = 1$ phases outside $r \approx 2$ a.u. and has the wrong phases inside. It cannot produce this back-scattering.

7.3 Interpretation of the tests

The diagnosis explains all the results of Section 6:

- The binary peak is correct in every version. It is determined by the initial state and the first-order projectile interaction, which are adequate.
- The recoil and large-angle deficit is the same for all three orbitals. It is a missing mechanism, not an orbital-specific problem.

- The short-range potential raises the large-angle TDCS isotropically, through the non-orthogonal overlap, not through physical back-scattering.
- Orthogonalization removes that artefact and leaves almost no back-scattering, because the phases of the ejected wave are still wrong.

7.4 Proposed improvement: distorted wave for the ejected electron

We propose to replace the 3CWZ ejected wave by the exact solution of Eq. (11) with the potential of Section 4:

$$\phi_e^-(\mathbf{r}) = \sqrt{\frac{2}{\pi}}\frac{1}{kr}\sum_l (2l+1)\, i^l\, e^{-i(\sigma_l+\delta_l)}\, u_l(r)\, P_l(\hat{\mathbf{k}}\cdot\hat{\mathbf{r}}), \tag{54}$$

where u_l is normalized to $\sin(kr - l\pi/2 + \eta \ln 2kr + \sigma_l + \delta_l)$.

Implementation. Only the numerical correction of the ejected wave changes. The structure of Eq. (30) is kept:

$$\phi_e\Phi \;\to\; \left[\phi_e^{DW} - \phi_e^{C}(Z=1)\right]\Phi + \phi_e^{C}(Z=1)\Phi. \tag{55}$$

- The second term is the existing analytic part.
- The first term is a finite sum over $l \le l_{\max} \approx 10$. Because the phase shifts persist at large r, it must be integrated over the extent of the orbital ($r \lesssim 12$ a.u.) instead of $r < d$.
- The projectile waves, the PCI factor and the Gamow factors are unchanged.

Consistency. The approach:

- uses exactly the potential of the variable-charge derivation;
- drops only the local approximation $\eta \to \eta(r)$ for the ejected electron;
- is the ejected-electron part of the distorted-wave methods [13, 12];
- follows the 1DW calculations for H_2O with Moccia orbitals [29].

Since the exact wave and the bound orbitals see the same potential, they are nearly orthogonal. The short-range projectile potential then becomes consistent with the ejected wave.

Planned checks:

- with all $\delta_l = 0$, the present $Z = 1$ results are recovered;
- the partial-wave sum of the $Z = 1$ Coulomb wave reproduces the code's wave on the grid;
- convergence in $l_{\max}$ and in the radial grid.

7.5 Domain of validity

Table 15 summarizes the domain of validity established in this work.

Table 15: Domain of validity of the molecular 3CWZ model (present implementation).

Reliable	Not reliable
Position and shape of the binary peak in asymmetric kinematics with small momentum transfer	Recoil region and large ejection angles (missing short-range distortion of the ejected electron)
Momentum profiles of EMS at the PWBA level (with converged grids)	Fast outgoing electrons or large momentum transfer ($K \gtrsim 3$ a.u.) with the default grids
Effective charges and target description (Moccia orbitals)	Integrated cross sections (DDCS, SDCS, total) before grid adaptation

To complete: Revise this section after the CH_4, PCI and constant-charge results.

8 Conclusions and perspectives

8.1 Conclusions

- **Formalism (O1).** The 3CWZ model was extended to molecular targets described by the one-centre SCF orbitals of Moccia:
  - the orientation average is analytic;
  - the effective charges follow from the monopole Hartree potential, with a self-interaction correction for the ejected electron;
  - the computed $Z(r)$ reproduces every feature expected from the derivation.
- **Code.** A validated parallel code treats any orbital of an XH_n molecule. It reproduces the original CH_4 code exactly, and its analytic integrals agree with numerical integration to 10^{-7}.
- **NH_3 TDCS (O2, O4).** For $2a_1$, 1e and $3a_1$ at $E_s = 500$ eV, $E_e = 74$ eV:
  - the binary peak is well reproduced;
  - the recoil peak and the large-angle side are 5–10 times too small.
- **Origin of the deficit.** The deficit comes from the ejected-electron wave. The local-charge Coulomb wave is not a solution of the one-electron equation in the screened ionic potential. It misses phase shifts of up to 2.6 rad and back-scattering about 100 times the Coulomb value. Neither the short-range projectile potential nor orthogonalization corrects it.
- **Additional tests.** The EMS profiles confirm the Moccia orbitals; the 1e high-momentum tail is slightly too small. The EMS and DDCS tests also delimit the numerical domain of the present grids.

To complete: Add the conclusions on CH_4 (O2, O4), on the PCI study (O3) and on the comparison with constant charge.

8.2 Perspectives

- Exact distorted wave for the ejected electron in the same Hartree potential, within the 3CWZ structure.
- Grids adapted to the momentum transfer, then DDCS, SDCS and total cross sections (comparison with Refs. [36, 40]).
- Exchange amplitude, and comparison with the low-energy data of Nixon *et al.* [41].
- Non-spherical ($L \geq 1$) terms of the potential, which matter for the dipolar NH_3.
- Extension to other molecules of interest (H_2O, biomolecular fragments) and to other geometries (perpendicular plane, low energy).

A Status of the protocol objectives

Objective / step	Realisation	Status
O1. Molecular 3CWZ with Moccia orbitals, $Z(r)$ per MO	Sections 3–5; code validated	Done
O2. TDCS for CH_4 and NH_3 orbitals	NH_3: three orbitals, El Mir, EMS, DDCS; CH_4: to do	Partial
O3. PCI with and without C_{12}	Section 6.7	To do
O4. Comparison with experiment	NH_3 vs [33, 35, 36]; CH_4 and [41, 40] to do	Partial
Contribution of the variable charge	$Z(r)$ derived and checked; constant-Z comparison to do	Partial
Domain of validity	Section 7	Partial
Bibliography (2 months)		Partial
OCE orbitals and $Z(r)$ (1 month)		Done
Molecular 3CWZ code (1 month)		Done
CH_4 calculations, PCI study (3 months)		To do
NH_3 calculations (3 months)		Partial
Analysis (2 months)		Partial

B Data and code

All files are in the repository `dramoros95/dd`, folder `nh3/`:

- `nh3_opt_claude.for`: the molecular 3CWZ code (options `isr`, `iorth`);
- `elmir_conditions/`: runs, digitized data and figures for Ref. [33];
- `rezkallah_conditions/`: EMS and DDCS tests, data and scripts;
- `diagnostics/`: $Z(r)$ and phase-shift programs;
- `theory_orthogonalization.md`, `theory_distorted_wave.md`: theory notes;
- `final_report/`: this report.

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